Inle Myat Bush

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Education

Cornell University 2021 - 2025
BA in Computer Science, Biology, and Mathematics

Experience

Han Lab, Molecular Biology and Genetics Department, Cornell 02/2022 - Present
Undergraduate Research Assistant

- Full-time summer intern designing and performing experiments in neurodegeneration and structural plasticity
- Lab skills: *Drosophila* husbandry; designing and performing genetic crosses ; immunohistochemistry; tissue dissection and mounting and confocal microscopy.

Navlakha Lab, CSHL 01/2022 - Present
Research Intern

- NSF funded summer position local nitrogen availability's effect on plant root development. Formalized plant root foraging in a reinforcement learning framework.
- Quantifying root branching structure dynamics and designing experiments to explore information processes and decision-making.

Su Lab, MCDB, CU Boulder 05/2019 - 08/2019
Research Intern

- Full time summer intern in genetics and cancer research, studying tumor culling mechanisms during development in *Drosophila melanogaster*.
- Coauthored paper published in *PLOS Genetics* (Recognized as second author).

Publications and Manuscripts

(In submission) Dasgupta S., Meirovitch Y., Zheng X., **Bush, I.**, Lichtman J., Navlakha S. (2023). A neural algorithm for computing bipartite matchings

(In submission) Xu Y., Wang B. **Bush, I.**, Saunders H.A., Wildonger J., Han C. (2023). Light-induced trapping of endogenous proteins reveals spatiotemporal roles of microtubule and kinesin-1 in dendrite patterning of Drosophila sensory neurons

Xu Y., **Bush, I.**, Han C. (2023). Visualization and Measurement of Dendrite Arborization in Neurons. *Neuronal Morphogenesis - Methods and Protocols*

Brown, J., **Bush, I.**, Bozon, J., & Su, T. T. (2020). Cells with loss-of-heterozygosity after exposure to ionizing radiation in Drosophila are culled by p53-dependent and p53-independent mechanisms. *PLOS Genetics*, 16(10). doi:10.1371/journal.pgen.1009056

Honors

NSF Scholar, Cold Spring Harbor Laboratory Undergraduate Research Program 2023
Grant Recipient, Einhorn Discovery Grant 2022
Best Overall, Cornell AppDev Hack Challenge 2021
Sponsor Prize, Cornell BRH Hackathon 2021
Dean's List, Cornell College of Arts and Sciences all semesters

Technical Skills

Languages: Ocaml, Julia, Python , MatLab, R, Java, C, C++, JavaScript, SQL, IJM, English, French, Burmese
Tools: Jupyter, VSCode, Eclipse, IntelliJ, RStudio, Flask, Maven, Postman, Git, ImageJ, Microsoft Office, LaTeX